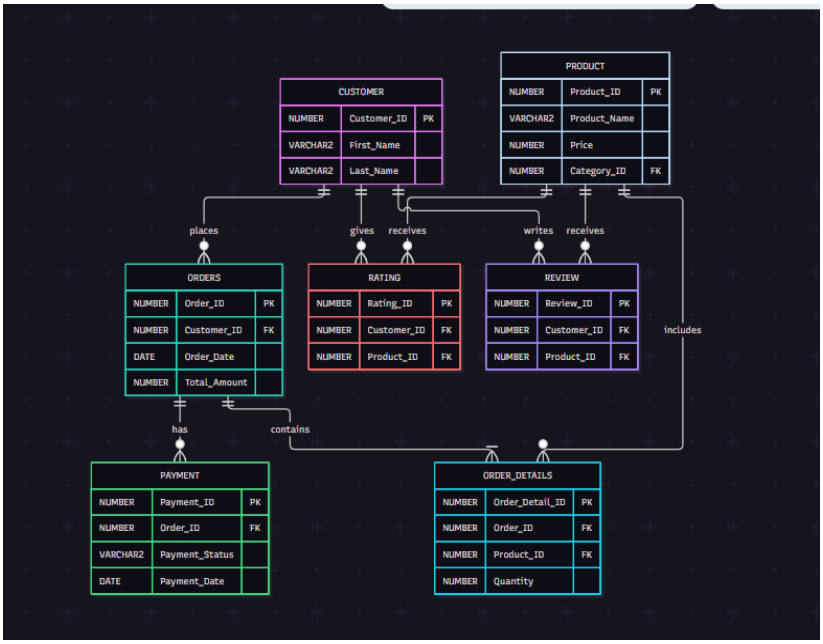
DATABASE RELATIONSHIP ANALYSIS USING JOINS

Pepperfry Furniture E-Commerce Database Management System

1. Combining Multiple Tables

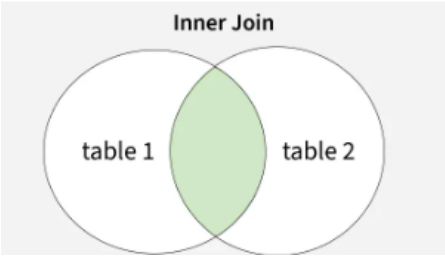
The ER Diagram represents the entities and relationships in the Pepperfry E-Commerce Database,

including Customer, Product, Orders, and Payment. It shows how tables are connected using Primary Keys and Foreign Keys to maintain data consistency.



2.INNER JOIN

INNER JOIN retrieves matching records from two or more tables based on a related column



2.1 Customer And Orders

INNER JOIN retrieves matching customer and order details using Customer_ID from both tables

```
SQL> SELECT C.Customer_ID,
  2         C.Customer_Name,
  3         O.Order_ID,
  4         O.Order_Date,
  5         O.Total_Amount
  6  FROM Customer C
  7  INNER JOIN Orders O
  8  ON C.Customer_ID = O.Customer_ID;

CUSTOMER_ID CUSTOMER_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT TOTAL_AMOUNT
--------- ------------
          1 Arun                                                        1
01-SEP-26        25000

          2 Priya                                                       2
02-SEP-26        18500

          3 Rahul                                                       3
03-SEP-26        32000


CUSTOMER_ID CUSTOMER_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT TOTAL_AMOUNT
--------- ------------
          4 Sneha                                                       4
04-SEP-26        12500

          5 Karthik                                                     5
05-SEP-26        45000
```

2.2 Orders and Payment

INNER JOIN retrieves matching order and payment details using the common Order_ID column

```
SQL> SELECT O.Order_ID,
  2         O.Order_Date,
  3         O.Total_Amount,
  4         P.Payment_ID,
  5         P.Payment_Method,
  6         P.Payment_Status,
  7         P.Amount
  8  FROM Orders O
  9  INNER JOIN Payment P
 10  ON O.Order_ID = P.Order_ID;

  ORDER_ID ORDER_DAT TOTAL_AMOUNT PAYMENT_ID PAYMENT_METHOD
---------- --------- ------------ ---------- --------------------
PAYMENT_STATUS           AMOUNT
-------------------- ----------
         1 01-SEP-26        25000          1 UPI
Completed                 15000

         2 02-SEP-26        18500          2 Credit Card
Completed                 25000

         3 03-SEP-26        32000          3 Debit Card
Pending                   12000


  ORDER_ID ORDER_DAT TOTAL_AMOUNT PAYMENT_ID PAYMENT_METHOD
---------- --------- ------------ ---------- --------------------
PAYMENT_STATUS           AMOUNT
-------------------- ----------
         4 04-SEP-26        12500          4 Cash
Completed                 18000
```

2.3 Product and Order_Details

INNER JOIN retrieves matching product and order details using the common Product_ID column from both tables

```
SQL> SELECT P.Product_ID,
  2         P.Product_Name,
  3         OD.Order_ID,
  4         OD.Quantity
  5  FROM Product P
  6  INNER JOIN Order_Details OD
  7  ON P.Product_ID = OD.Product_ID;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
  ORDER_ID   QUANTITY
---------- ----------
         1
Sofa
         1          4

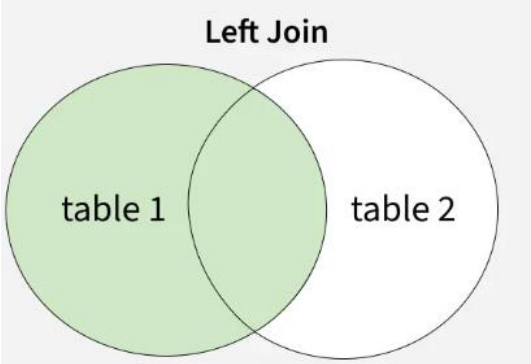
         2
Dining Table
         2          1

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
  ORDER_ID   QUANTITY
---------- ----------

         3
Office Chair
         3          2
```

3. LEFT JOIN

LEFT JOIN returns all records from the left table and matching records from the right table. If there is no match, the right table's columns return NULL values



3.1 Display All Customers and Their Orders

LEFT JOIN displays all customers and their order details

```
SQL> SELECT C.Customer_ID,
  2         C.Customer_Name,
  3         O.Order_ID,
  4         O.Order_Date,
  5         O.Total_Amount
  6  FROM Customer C
  7  LEFT JOIN Orders O
  8  ON C.Customer_ID = O.Customer_ID;

CUSTOMER_ID CUSTOMER_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT TOTAL_AMOUNT
--------- ------------
          1 Arun                                                        1
01-SEP-26        25000

          2 Priya                                                       2
02-SEP-26        18500

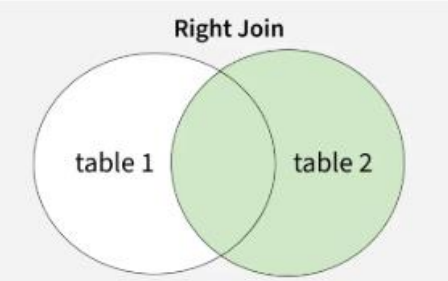
          3 Rahul                                                       3
03-SEP-26        32000


CUSTOMER_ID CUSTOMER_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT TOTAL_AMOUNT
--------- ------------
          4 Sneha                                                       4
04-SEP-26        12500

          5 Karthik                                                     5
05-SEP-26        45000
```

4. RIGHT JOIN

RIGHT JOIN returns all records from the right table and matching records from the left table. If there is no match, the left table's columns return NULL values



4.1 Display All Orders and Customer Details

RIGHT JOIN displays all orders and their matching customer details

```
SQL> SELECT C.Customer_ID,
  2         C.Customer_Name,
  3         O.Order_ID,
  4         O.Order_Date,
  5         O.Total_Amount
  6  FROM Customer C
  7  RIGHT JOIN Orders O
  8  ON C.Customer_ID = O.Customer_ID;

CUSTOMER_ID CUSTOMER_NAME                                     ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT TOTAL_AMOUNT
--------- ------------
          1 Arun                                                       1
01-SEP-26        25000

          2 Priya                                                      2
02-SEP-26        18500

          3 Rahul                                                      3
03-SEP-26        32000


CUSTOMER_ID CUSTOMER_NAME                                     ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT TOTAL_AMOUNT
--------- ------------
          4 Sneha                                                      4
04-SEP-26        12500

          5 Karthik                                                    5
05-SEP-26        45000
```

5. Complete Order Details

This query retrieves complete order details, including customer information, product details, quantity, order status, and payment information by joining multiple tables

```
SQL> SELECT O.Order_ID,
  2  O.Order_Date,
  3  O.Total_Amount,
  4  C.Customer_ID,
  5  C.Customer_Name,
  6  P.Product_ID,
  7  P.Product_Name,
  8  OD.Quantity,
  9  PM.Payment_Method,
 10  PM.Payment_Status
 11  FROM Orders O
 12  INNER JOIN Customer C
 13  ON O.Customer_ID = C.Customer_ID
 14  INNER JOIN Order_Details OD
 15  ON O.Order_ID = OD.Order_ID
 16  INNER JOIN Product P
 17  ON OD.Product_ID = P.Product_ID
 18  LEFT JOIN Payment PM
 19  ON O.Order_ID = PM.Order_ID;

  ORDER_ID ORDER_DAT TOTAL_AMOUNT CUSTOMER_ID
---------- --------- ------------ -----------
CUSTOMER_NAME                                      PRODUCT_ID
-------------------------------------------------- ----------
PRODUCT_NAME
--------------------------------------------------------------------
  QUANTITY PAYMENT_METHOD                 PAYMENT_STATUS
---------- ------------------------------ --------------------
         1 01-SEP-26        25000           1
Arun                                                         1
Sofa
         4 UPI                            Completed
```

6. Customer Purchase History

```
SQL> SELECT C.Customer_ID,
  2  C.Customer_Name,
  3  O.Order_ID,
  4  O.Order_Date,
  5  P.Product_Name,
  6  OD.Quantity,
  7  O.Total_Amount
  8  FROM Customer C
  9  INNER JOIN Orders O
 10  ON C.Customer_ID = O.Customer_ID
 11  INNER JOIN Order_Details OD
 12  ON O.Order_ID = OD.Order_ID
 13  INNER JOIN Product P
 14  ON OD.Product_ID = P.Product_ID
 15  ORDER BY C.Customer_ID;

CUSTOMER_ID CUSTOMER_NAME                                    ORDER_ID
----------- ---------------------------------------------- ----------
ORDER_DAT
---------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY TOTAL_AMOUNT
---------- ------------
         1 Arun                                                    1
01-SEP-26
Sofa
         4        25000
```

7. Customer Purchase History with Payment

This query displays customer purchase history with product details, order amounts, and payment methods and statuses

```
SQL> SELECT C.Customer_ID,
  2  C.Customer_Name,
  3  O.Order_ID,
  4  O.Order_Date,
  5  P.Product_Name,
  6  OD.Quantity,
  7  O.Total_Amount,
  8  PM.Payment_Method,
  9  PM.Payment_Status
 10  FROM Customer C
 11  INNER JOIN Orders O
 12  ON C.Customer_ID = O.Customer_ID
 13  INNER JOIN Order_Details OD
 14  ON O.Order_ID = OD.Order_ID
 15  INNER JOIN Product P
 16  ON OD.Product_ID = P.Product_ID
 17  LEFT JOIN Payment PM
 18  ON O.Order_ID = PM.Order_ID
 19  ORDER BY C.Customer_ID;

CUSTOMER_ID CUSTOMER_NAME                                    ORDER_ID
----------- ---------------------------------------------- ----------
ORDER_DAT
---------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY TOTAL_AMOUNT PAYMENT_METHOD                 PAYMENT_STATUS
---------- ------------ ------------------------------ --------------------
         1 Arun                                                    1
01-SEP-26
Sofa
         4        25000 UPI                            Completed
```

8. Multi-Table Business Report

```
SQL> SELECT
  2  C.Customer_Name,
  3  COUNT(DISTINCT O.Order_ID) AS Total_Orders,
  4  SUM(OD.Quantity * P.Price) AS Total_Purchase,
  5  SUM(OD.Quantity) AS Total_Products
  6  FROM Customer C
  7  INNER JOIN Orders O
  8  ON C.Customer_ID = O.Customer_ID
  9  INNER JOIN Order_Details OD
 10  ON O.Order_ID = OD.Order_ID
 11  INNER JOIN Product P
 12  ON OD.Product_ID = P.Product_ID
 13  GROUP BY C.Customer_Name
 14  ORDER BY Total_Purchase DESC;

CUSTOMER_NAME                                      TOTAL_ORDERS TOTAL_PURCHASE
-------------------------------------------------- ------------ --------------
TOTAL_PRODUCTS
--------------
Arun                                                          1         108000
             4

Vijay                                                         1          44000
             2

Sneha                                                         1          30000
             1
```

9. Payment Transaction Report

This query generates a payment transaction report using JOIN to display order details, customer names, payment methods, payment statuses, and payment amounts

```
SQL> SELECT
  2  O.Order_ID,
  3  C.Customer_Name,
  4  O.Total_Amount,
  5  P.Payment_ID,
  6  P.Payment_Method,
  7  P.Payment_Status,
  8  P.Amount
  9  FROM Customer C
 10  INNER JOIN Orders O
 11  ON C.Customer_ID = O.Customer_ID
 12  INNER JOIN Payment P
 13  ON O.Order_ID = P.Order_ID
 14  ORDER BY O.Order_ID;

  ORDER_ID CUSTOMER_NAME                                      TOTAL_AMOUNT
---------- -------------------------------------------------- ------------
PAYMENT_ID PAYMENT_METHOD                 PAYMENT_STATUS           AMOUNT
---------- ------------------------------ -------------------- ----------
         1 Arun                                                      25000
         1 UPI                            Completed                15000

         2 Priya                                                     18500
         2 Credit Card                    Completed                25000

         3 Rahul                                                     32000
         3 Debit Card                     Pending                  12000
```